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*Université
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Ecole
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Mobile Robot for Road Aging Analysis in Togo

Presented by the Robotics Club of the Polytechnic School of Lomé, Academic Year 2024-2025.

This project proposes an autonomous and efficient robotic solution to revolutionize the monitoring and analysis of road infrastructure aging in Togo.



Some Damaged Roads in Togo



The Urgency of Road Degradation in Togo



Key Factors

Tropical climate, vehicle overloads, and aging materials.



Financial Impact

High repair costs, estimated at 15-20% of the national road budget.

Togolese roads are undergoing accelerated degradation. This leads to significant costs and risks.



Limitations of Current Manual Inspection

Limited Frequency

Rare inspections (1-2 times per year). This does not allow for continuous monitoring.

Time-Consuming Process

Manual methods are slow and inefficient. They delay maintenance.

Human Errors

Inspections are subject to inaccuracies. This leads to inconsistent data.

Manual inspection is insufficient for our roads. It lacks precision and regularity.





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The Opportunity of Mobile Robotics

Autonomous Analysis

The robot offers monitoring without constant human intervention. This improves efficiency.

Frequency and Precision

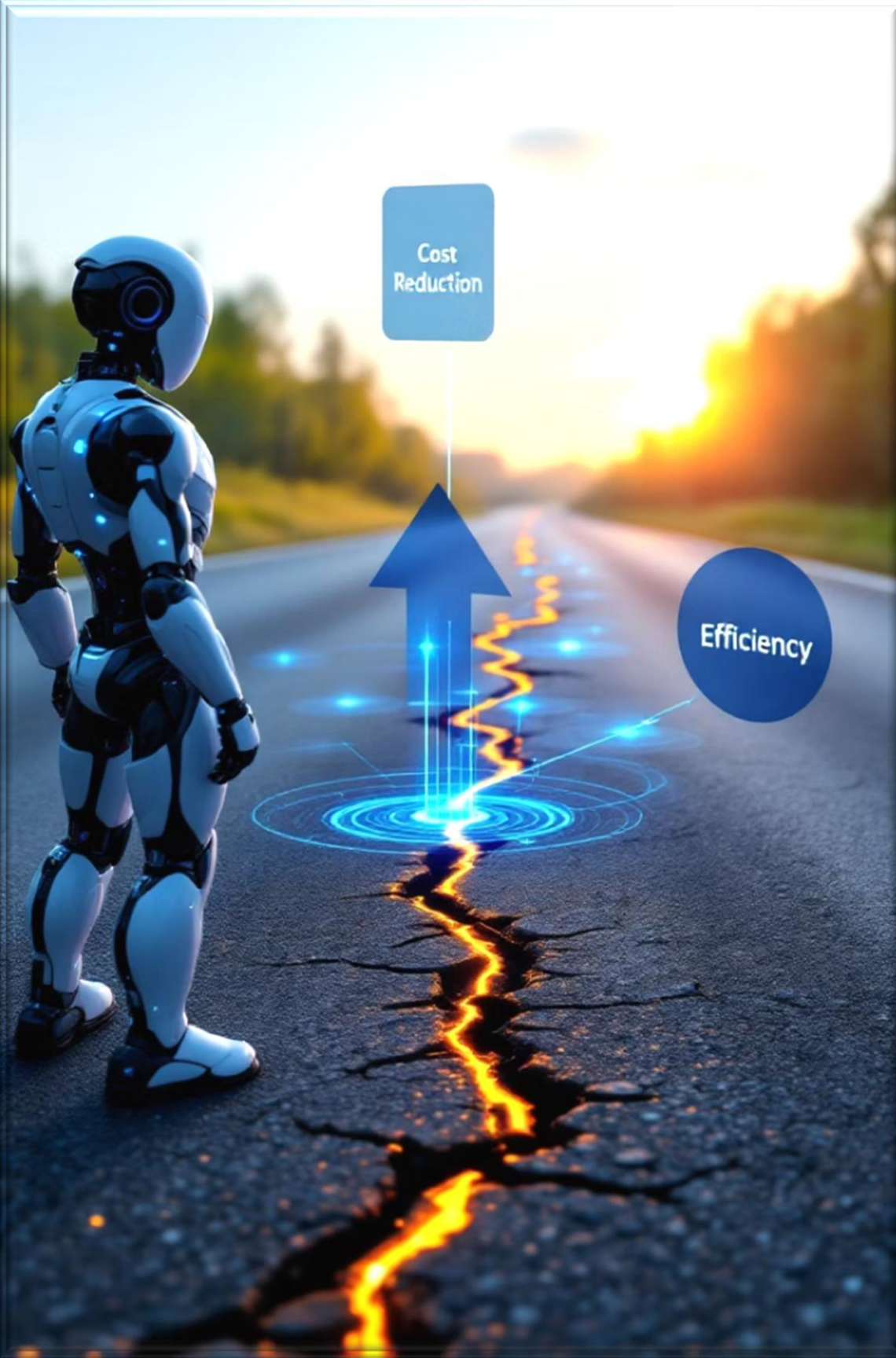
Regular inspections and accurate data are possible. This optimizes maintenance.

Cost Reduction

The project aims to reduce maintenance expenses. It contributes to road safety.

Mobile robotics represents an innovative solution. It allows for proactive management of road infrastructure.





Project Problematic: Utility and Technological Choice

1

Project Utility

Design an autonomous and economical tool. It monitors the condition of Togolese roads.

2

Technological Choice

Prefer a terrestrial robot over a drone. For reliable and precise surface analysis.

Our project answers two fundamental questions. How to build an effective solution and why a terrestrial robot is the best choice.





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Why This Project Is Useful for Togo



Cost-Effective Prevention

Reduces maintenance costs by 20-30% due to early detection of problems.



Improved Safety

Decreases road accidents by approximately 10% related to degraded roads.



Local Adaptability

Affordable and locally maintainable solution, ensuring its sustainability.

This project offers significant benefits for Togo. It ensures safer roads and more economical management.



Why a Terrestrial Robot Rather Than a Drone

1 Direct Contact

Precise analysis of cracks and roughness.
Drones cannot offer this.

2 Resilience

Works in rain or wind. Drones are limited by weather conditions.

3 Regulations

Avoids overflight restrictions. Drone regulations are often complex.

4 Autonomy

Long-lasting batteries (up to 8h). This ensures prolonged operations.

The terrestrial robot surpasses the drone for this application. It offers superior reliability and autonomy.



Mobile Robot Project Objectives

Main Objective

Develop an autonomous mobile robot to analyze the condition of Togolese roads.

1

2

Robust Design

Mechanics adapted to rough terrain.

3

Sensor Integration

LIDAR, cameras, vibration sensors for surface analysis.

4

Control and Data

Mobile application via Raspberry Pi 5. Data transmission to database (MySQL/MongoDB).

5

Local Resilience

Withstands dust, heat (35-40°C), and intense rain.

Our objectives are clear and ambitious. We want a high-performance robot adapted to the Togolese context.



Key Phases of Robot Development

1

Preliminary Research

Study of existing solutions. Definition of technical specifications.

2

Prototyping and Testing

Construction of the first prototype. Laboratory and simulated field tests.

3

Pilot Deployment

Robot trials on real road sections in Togo. Initial data collection.

4

Continuous Optimization

System improvement based on feedback. Integration of new functionalities.

The development will follow a structured approach. Each phase is essential for the project's success.





Expected Impact and Next Steps

30%

Cost Reduction

Potential savings on road maintenance.

10%

Accident Decrease

Improvement of road safety.

1

Unique Solution

An autonomous mobile robot in Togo.

We are confident in the positive impact of this project. It will contribute to the development of Togolese infrastructure.

Next steps: Finalization of the design and start of prototype construction.



Robot Technical Presentation

Robot Configuration

Details on the physical structure of the robot, its hardware components, and their specific arrangement for road analysis.

Kinematics

Explanation of the robot's movements, its joints, and how it navigates different types of road terrain.

Control via Application

Description of the mobile user interface used to control the robot, view real-time data, and manage missions.

Analysis Section

Presentation of the algorithms and methods used to process data collected by sensors and identify road degradations.



Robot Configuration

The robot is designed with robust and precise components to ensure reliable analysis of Togolese roads.



Robust Chassis

Durable and resistant structure, adapted to difficult terrains and varied climatic conditions. The use of aluminum or reinforced plastic is preferred.



Locomotion System

Equipped with rubber suspensions (for shocks) and fixed wheels for omnidirectional movement to allow stable and efficient navigation on different types of road surfaces. (They will later be replaced by Mecanum wheels).



Integrated Sensors

Includes high-resolution cameras, a LiDAR, and position sensors (GPS, IMU) for navigation.



Sensors



Ultrasonic Sensor (HC-SR04)

Role: Measures the depth of ruts and detects obstacles, ensuring precise navigation and detailed data collection on road surface irregularities.



Accelerometer/Gyroscope (MPU6050)

Role: Analyzes vibrations induced by traffic and repeated loads, providing essential information on road dynamics and structural constraints.



Strain Gauges (HX711 + 350 Ω Gauge)

Role: Measures micro-deformations in wearing courses, allowing for the detection of early signs of fatigue and degradation of the road pavement.



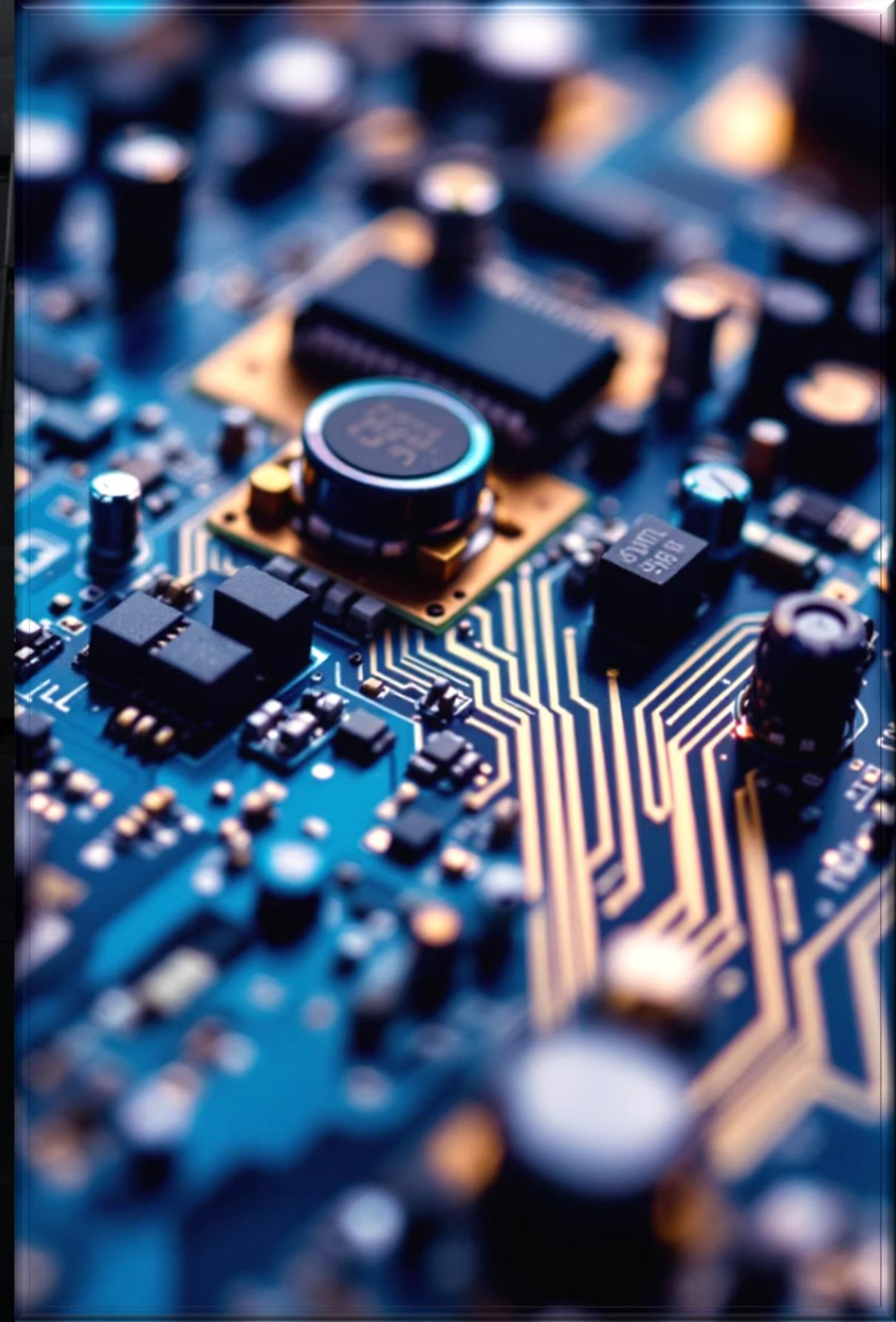
Onboard Electronics

Motor Control

- **Main Control Board:** An Arduino Mega 2560 precisely manages motors and PID servo control for smooth navigation.
- **Motor Drivers:** H-bridges (L298N or TB6612FNG) provide the necessary power, supporting 12V and 2A per channel.

Computation and Communication

- **Onboard Mini-computer:** A Raspberry Pi 4 is used for analyzing collected data and ensures communication with the central database.



Kinematics

Forward Kinematics

Forward kinematics involves calculating the robot's current position and orientation in space from the velocities or positions of its joints, particularly the wheel speeds. It allows us to know where the robot is and in what direction it is moving, given the commands sent to its motors.

Inverse Kinematics

Inverse kinematics (or indirect kinematics) is the inverse process: it determines the velocities or positions of the joints (the robot's wheels) required to reach a specific target position and orientation. This calculation is essential for trajectory planning and motion control, allowing the robot to navigate to a desired point.



Robot System Configuration

- **Structure:** Four fixed (non-steerable) wheels for stable locomotion.
- **Motorization:** Each wheel is independently driven by its own motor.
- **Reference Frames:** I is the global (inertial) reference frame of the environment, and B is the reference frame centered on the robot.
- **Geometric and Motion Parameters:** The robot's position is defined by (x, y) , its orientation by ψ . Its local velocities are u (forward), v (lateral), and ω (angular rotation).



Forward Kinematics

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = J(\psi) \begin{bmatrix} u \\ v \\ \omega \end{bmatrix}$$

Description

Transformation of velocities expressed in the robot's local frame to the global frame

Utility

Determine the robot's global speed from local speed commands



Inverse Kinematics

$$\begin{bmatrix} u \\ v \\ \omega \end{bmatrix} = J^{-1}(\psi) \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix}$$

Description

Transformation of velocities expressed in the global frame to the robot's local frame.

Utility

Determine the local velocity commands required to achieve a target expressed in the global frame.



Degrees of Freedom

- 1

 Translation along X-axis
- 2

 Translation along Y-axis
- 3

 Rotation around Z-axis

Hypothesis: the robot moves on a non-slippery surface



Wheel Kinematics Associated with the Robot

- **Direct Kinematics:** A fixed wheel rolls in its longitudinal direction. Taking x_{ci} as the speed and r as the radius

$$x_{ci} = (u - \omega d_{yi}) \cos \phi_i + (v + \omega d_{xi}) \sin \phi_i \quad \rho_i = \frac{1}{r} x_{ci}$$

- for $\phi_i = 0$ and wheel 1: (L, W); wheel 2: (L, -W); wheel 3: (-L, W), wheel 4: (-L, -W)

$$x_{ci} = u - \omega d_{yi}$$

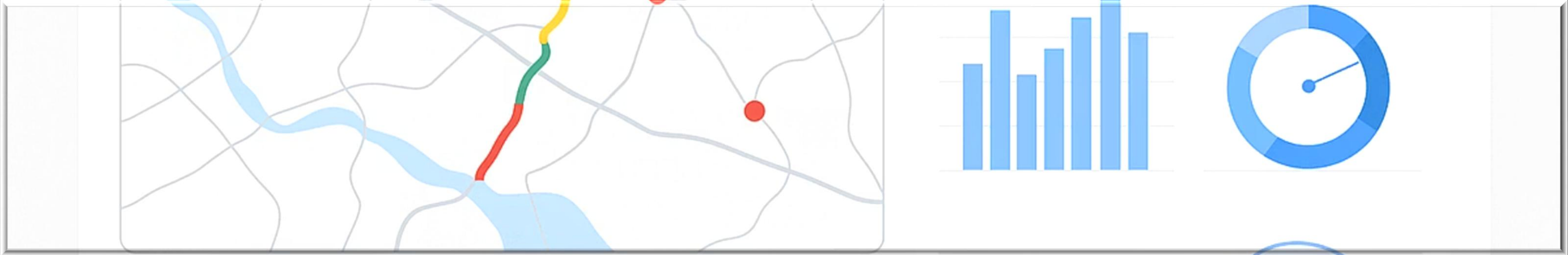
Using these positions, the value of d_{yi} varies between W and $-W$. The angular velocity of each wheel (ρ_i) is then given by:

$$\rho_i = \frac{1}{r} (u \pm \omega W)$$

- Inverse Kinematics

$$\begin{aligned} u &= \frac{r}{2} (\rho_1 + \rho_2) \\ \omega &= \frac{r}{2W} (\rho_2 - \rho_1) \\ v &= 0 \end{aligned}$$





Robot Interface

Interactive Map

- Display of the **robot's path** on a geographical map.
- Localization of **degraded areas** with a color code according to severity.

Data Visualization

- **Curves** (based on date, temperature, humidity, vibrations, etc.).
- **Radar graphs** for general road segment conditions.

Alerts and Recommendations

- Automatic alert system in case of critical degradation.
- Intervention suggestions: **urgent, medium, or to monitor.**

Data Export

- Export analysis reports in **PDF, Excel, or JSON** formats.
- Possible integration with GIS (Geographic Information Systems).



Data Analysis

Analysis Objectives

- Automatically assess road conditions.
- Identify critical road areas.

Key Indicators Extracted

- Cracking rate per square meter.
- International Roughness Index (IRI).
- Road Health Score (scale from 0 to 100).



Budget plan

Sensors	\$ 600
Robot physical components (onboard electronics, mechanics)	\$ 200
Miscellaneous (software development, data analysis, testing and deployment)	\$ 200
Total budget	\$ 1000





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Conclusion & Outlook

This project represents an essential technological response to the road infrastructure challenges in Togo. Its adaptable and modular approach offers potential for progressive expansion to other African countries facing similar issues. Ultimately, it could become the foundation of a regional predictive road maintenance strategy, relying on data analysis and automation to optimize road network management.

